SERIN MIN

Travel Behavior Modeling · Agent-Based Simulation · Digital Twins 3ody9@uos.ac.kr

Education

University of Seoul Mar 2024 – Feb 2026
Master of Science in Smart City (4.5/4.5) *Seoul, Korea*

- Transportation Planning Laboratory, Department of Transportation Engineering; Advisor: Prof. Seungjae Lee
- Thesis: Development of a Digital Twin System for Pedestrian Dynamics and Congestion Assessment in Transit Hubs

Kyung Hee University Mar 2019 – Feb 2024
Bachelor of Science with Double Majors in Landscape Architecture and Geography (4.34/4.5) *Seoul, Korea*

Work Experience

MIMLAB (Mobility & Intelligence Modeling Laboratory) Mar 2026 – Present
Data Scientist / Associate Researcher *Seoul, Korea*

- Develop **C++-based microscopic pedestrian behavior models** extending the Social Force Model with route choice, adaptive responses, and perceived stress, and use the simulator to study behavior under changing demand and spatial conditions.
- Developed a **Spatial LLM multi-agent framework** linking relational urban data with large-scale agent deliberation to represent heterogeneous citizen responses and evaluate alternative urban planning scenarios.

Transportation Planning Lab, University of Seoul Jan 2024 – Feb 2026
Researcher *Seoul, Korea*

- Performed **travel demand forecasting and network assignment** in EMME for 14 road and rail infrastructure projects, evaluating demand shifts and network impacts across planning scenarios and conducting benefit–cost analyses.
- Developed **UAV-derived 3D reference maps** for autonomous localization, including spatial feature extraction, map reduction, and field validation with autonomous mobility systems.

LPSCAPE Jan 2022 – Jun 2022, Jan 2023 – Feb 2023
Urban Planner / Intern *Seoul, Korea*

- Developed TOD and mobility strategies for urban development projects with a focus on transit accessibility.

Publications (A: article, C: conference presentation, *: equal contribution)

3 peer-reviewed journal articles · 11 conference presentations

[A3] **Infrastructure-led urban development does not have to come at the expense of environmental sustainability**
M Choi*, S Min*, S Kim, A Cho, M Bencekri, Y V Fan, B E Balassa, S Lee *Cities*, 2026

[A2] Network construction strategies for the activation of autonomous mobile robots
S Park, S Min, M Choi, S Kim, S Lee *Proceedings of the Institution of Civil Engineers – Municipal Engineer*, 2026

[C9] Whose city is faster? Invisible mobility and distributional equity in Seoul's GTX implementation
S Min, S Lee *International Conference on Urban Affairs (ICUA)*, Chicago, IL, USA, 2026

[C8] Pedestrian network reconfiguration in Seoul through street space reallocation
S Min, S Lee *American Association of Geographers (AAG) Annual Meeting*, San Francisco, CA, USA, 2026

[A1] **Autonomous vehicle integration with public transit for congestion mitigation and energy efficiency**
M Choi, S Min, J Kim, S Kim, J Kwak, S Lee *Sustainable Energy Technologies and Assessments*, 2025

[C5] **AI-augmented digital twin: Real-time pedestrian simulation in dynamic urban spaces**
S Min, M Choi, S Lee *Computational Urban Planning and Urban Management (CUPUM)*, London, UK, 2025

Manuscripts in Progress

From urban-rail space filling to metropolitan coverage: Diagnosing growth regimes in Seoul's railway network, 1984–2023
S Kim, S Min, I Jeong, S Lee Major revision, *Transportation Research Part A: Policy and Practice*

Spatial reach and station access in metropolitan rail networks: A scaling-based typology from twenty city-regions worldwide
S Kim, S Min, J Kwak, J Lee, S Lee Major revision, *Journal of Transport Geography*

Quantifying nonlinear congestion transitions in transit networks via agent-based pedestrian simulation
S Min, S Lee To be submitted to *Computers, Environment and Urban Systems*

Sparse reference 3D map conditioning for resource-constrained mobile robot localization in GNSS-degraded environments
M Choi*, S Min*, S Park, S Lee, S Lee To be submitted to *IEEE Transactions on Intelligent Transportation Systems*

Patents

[P2] **Language Model and MaaS-based Traffic Recommendation System and Method Thereof** Patent No. 10-2865202

[P1] **System for Automatic Generation of Fixed Object Reference Points and Convex Hulls Using Remote Sensing Data and Method Thereof** Patent No. 10-2840859

Selected Research & Applied Projects

Human-Calibrated Pedestrian Behavior Modeling in Super-Aged Urban Environments 2026
Architecture & Urban Research Institute (AURI)

- Calibrated and validated microscopic pedestrian behavior models using living-lab and VR data from 113 participants to quantify how environmental barriers and emerging mobility modes affect movement, conflict, and perceived stress.

Digital Twin Simulation for GITC Multimodal Transit Operations at Samseong Station 2026

- Developed a pedestrian digital twin combining route-choice logic, visibility-aware Social Force models, and perceived-stress measures to examine pedestrian behavior and congestion under alternative demand, circulation, and capacity scenarios.

Spatial LLM and Multi-Agent Urban Planning 2026
Architecture & Urban Research Institute (AURI)

- Constructed a relational spatial network of roads, buildings, parcels, and access points to support LLM-based spatial reasoning and multi-agent evaluation of alternative planning scenarios.

Travel Demand and Network Response to Road Space Reallocation in Seoul 2025 – 2026
Seoul Metropolitan Government & The Seoul Institute

- Prioritized pedestrian-improvement corridors using entropy-weighted TOPSIS, then combined EMME demand reassignment with SUMO microsimulation to quantify demand redistribution and network-wide effects of lane and roadway reallocation.

Sparse 3D Reference Mapping for GNSS-Degraded Autonomous Localization 2024 – 2026
Ministry of Land, Infrastructure and Transport (MOLIT) & KAIA

- Compressed UAV-derived point clouds into sparse object-based reference maps for auxiliary localization, cutting computational load by 83% while maintaining positioning accuracy in real-world autonomous mobility experiments.

Awards & Honors

Outstanding Oral Presentation Award Sep 2025
Awarded for UAV imagery and deep learning-based automated road crack detection research · Korean Society of Transportation (KST).

Excellence Award, Pseudonymized Data Utilization Competition Sep 2025
Awarded for improving travel behavior prediction using linked pseudonymized telecom data · Ministry of Science and ICT & PIPC.

Outstanding Oral Presentation Award Oct 2024
Awarded for Urban Network Analysis-based gated community walkability research · Urban Design Institute of Korea (UDIK).

Outstanding Graduate Award Feb 2024
Selected as the department's sole recipient · Korean Institute of Landscape Architecture (KILA).

Grand Prize, Digital Twin Technology Application Competition Dec 2023
Awarded for UAV-based 3D reconstruction and digital twin modeling using remote sensing data · Kyung Hee University.

Bronze Prize (Finalist), The 20th National Exhibition of Korean Landscape Architecture Sep 2023
Awarded for landscape architecture design addressing urban and environmental planning challenges · KILA.

Scholarships & Fellowships

Graduate Academic Excellence Scholarship (4 Semesters) 2024 – 2026
University of Seoul

BK21 Research Fellowship 2024 – 2026
National Research Foundation of Korea (NRF) / University of Seoul

Undergraduate Academic Excellence Scholarship (6 Semesters) 2019 – 2024
Kyung Hee University

Specialized Skills

Programming Languages: C++, Python, R, MATLAB, Java

Transportation Modeling & GIS: TransCAD, SUMO, EMME, ArcGIS, QGIS

3D Modeling & Mapping: AutoCAD, Rhino, Unreal Engine, Unity, DJI, Pix4D, CloudCompare

Design & Typesetting: Adobe Photoshop, Illustrator, InDesign, LaTeX